PSYC100: General Psychology

Week 3: The Predictably Irrational Mind

Milestone 1: Cognitive Shortcuts & Attention Limits

The human brain operates as a high-efficiency shortcut machine. To manage an overwhelming influx of environmental data under strict time constraints, the mind relies on baseline intuitive rules known as heuristics, sacrificing absolute precision to conserve mental energy.

Lesson 1: The Fast-Triaging Mind & Core Heuristics

Heuristics allow for rapid, automatic decisions that are usually 'good enough' for immediate survival but systematically introduce predictable cognitive biases.

■ Expanded Content: Core Shortcuts Unpacked

- **Representativeness Heuristic:** Judging probability based on how closely an item matches an internal stereotype, while completely ignoring baseline mathematical odds.
- **Availability Heuristic:** Estimating occurrence frequency based entirely on how easily or vividly examples can be recalled from memory paths.
- **Inattentional Blindness:** A severe failure of selective attention where an individual misses an obvious, visible stimulus because focus is fully consumed.

```
# Practical Application: Triage Heuristic vs. Exhaustive Validation
def fast_triage(patient_profile):
    if patient_profile.get("clutching_chest") and patient_profile.get("sweating"):
        return "Priority 1: Immediate Cardiovascular Intervention"
    return "Priority 2: Standard Assessment Queue"
```

Milestone 2: Asymmetrical Risks & Anchoring Mechanics

Human choices are rarely absolute; they are highly dependent on starting reference points and how options are mathematically or emotionally framed.

Lessons 2 & 3: Anchoring, Framing, and Prospect Theory

Tversky & Kahneman demonstrated that initial numbers set an unconscious anchor, pulling downstream calculations toward it. Furthermore, prospect theory proves our value curve is asymmetrical: the pain of a loss hurts roughly twice as much as an equivalent gain feels good.

```
# Simulation: Reference Point Shifts & Diminishing Sensitivity
def evaluate_prospect_value(amount, reference_point):
    net_value = amount - reference_point
    if net_value >= 0:
        return net_value ** 0.5 # Concave curve for gains
    else:
        return -2.0 * (abs(net_value) ** 0.5) # Steeper convex loss curve
```

Lesson 4: Reconstructive Memory & False Memories

Memory does not function like a pristine video recording replay. Instead, it operates as an active, post-event reconstruction that can be easily contaminated or manipulated by external suggestions.

■ Expanded Content: The Loftus Car-Crash Experiment Mechanics

- **Leading Question Effect:** Changing a single verb in a post-event query (asking how fast cars went when they 'smashed' vs. 'hit' each other) shifted speed reconstructions significantly (40.8 mph vs. 34.0 mph) despite participants watching identical footage.
- **False Memory Injection:** The word 'smashed' acted as a leading context clue, causing brains to generate logical puzzle pieces during reconstruction. This led participants to confidently 'remember' details that never occurred, such as non-existent broken glass.
- **Calibration Tactic:** To counteract reconstructive bias, systematically review objective facts over personal assumptions, relying on hard records, timestamps, or documentation rather than immediate cognitive recall.

```
# Mock Application: Reconstructive Memory Trace Distortion Simulation
def reconstruct_event_memory(base_facts, leading_verb):
    memory_trace = dict(base_facts)
    if leading_verb == "smashed":
        memory_trace["estimated_speed_mph"] = 40.8
        memory_trace["recalled_broken_glass"] = True
    elif leading_verb == "hit":
        memory_trace["estimated_speed_mph"] = 34.0
        memory_trace["recalled_broken_glass"] = False
    return memory_trace
```

Milestone 3: Systemic Calibration & Protection Matrices

True professional execution requires establishing operational safeguards that prevent silent cognitive distortions from hijacking data review pipelines.

Lessons 5–10: Metacognitive Gaps & Behavioral Guardrails

The Dunning-Kruger effect highlights a dual burden: incompetent execution is paired with a metacognitive blind spot that prevents self-correction. Calibrating judgments requires strict process guardrails.

Cognitive Distortion	Interpersonal & Technical Risk	Systemic Solution / Guardrail
The Halo Effect	Past presentation skill skews asset technical reviews.	`Review each asset completely independently using blind criteria.`
Inattentional Blindness	Tunnel focus causes critical alerts to go completely unseen.	`Enforce a strict 'Scan Before Submit' top-to-bottom screening ritual.`
The Marshmallow Test	Immediate hot-system impulses deplete limited willpower.	`Engineer environment: turn on Do Not Disturb for fixed 30-min blocks.`
Dunning-Kruger Trap	Overconfidence leading to severe task underestimation.	`Calibrate self-assessments by tracking actual times vs. estimates.`
Somatic Marker Defect	Ignoring useful physiological body warnings.	`Incorporate markers: pause when a traffic/code scenario 'feels off.'`

```
# Operational Safety: Multi-Variant Safeguard Pipeline
def execute_sprint_planning(initial_estimate_months, independent_peer_reviews):
    if len(independent_peer_reviews) == 0:
        return f"REJECTED: Anchoring risk high on initial {initial_estimate_months} month value."
    calibrated_average = sum(independent_peer_reviews) / len(independent_peer_reviews)
    return f"Calibrated Project Length Verified: {calibrated_average} months."
```